Generated report 1: pbmc

Objective (as given)

"Learn a low-dimensional representation and generate realistic synthetic cells for sharing"

Agent's decision

Task: representation_learning | decision: proceed | task run: representation_learning | also: data_generation

profile.json reports 2,700 cells (well above the 200-sample floor) with no missing values, and neither task needs labels (has_labels is false, which does not matter here). The data are non-negative integer UMI counts (is_nonneg_integer true) with 97.4% zeros, so a count-likelihood generative model (negative-binomial VAE) is a sound way to get both a latent representation and samples. A non-generative alternative (PCA on log-normalized data) would give a low-dimensional representation but cannot generate count-valued cells, so it is not a substitute for the full objective; PCA is still the evaluation space and the flow comparator is built on it. DATA_DESCRIPTION.md warns that with only 2,700 cells large models can memorize training cells, so memorization checks against held-out cells are part of the success criterion.

Models run (headline metrics; details on the metrics page)

Run names are <model id>_s<seed>: vae_nb_s0 is model vae_nb trained with seed 0. Models in this report: vae_nb (seeds 0, 1); flow_pca (seeds 0, 1). The next page explains seeds, the evaluation space, references, and memorization.

```
vae_nb_s0          vae    C2ST 0.9457 (real-vs-real 0.5855)  precision 0.7926  recall 0.7667  DCR ratio 1.4897
vae_nb_s1          vae    C2ST 0.9253 (real-vs-real 0.5664)  precision 0.8722  recall 0.7667  DCR ratio 1.4131
flow_pca_s0        flow   C2ST 0.8572 (real-vs-real 0.5855)  precision 0.5611  recall 0.987  DCR ratio 1.5933
flow_pca_s1        flow   C2ST 0.8352 (real-vs-real 0.5664)  precision 0.5315  recall 0.9833  DCR ratio 1.5531
```

Downstream-use verdicts

- [conditional] Use the NB-VAE 10-dimensional latent codes as a low-dimensional representation of these PBMC cells (input to clustering, neighbor graphs, visualization) (vae_nb_s0)

- [not_recommended] Use NB-VAE synthetic cells as a stand-in for real PBMC cells in analyses that depend on the joint gene-expression distribution (clustering, cell-population structure, differential expression) (vae_nb_s0)

- [conditional] Use NB-VAE synthetic count matrices where only per-gene and per-cell marginal properties matter (e.g. testing a processing pipeline, teaching, method demos on count-shaped data) (vae_nb_s0)

- [not_recommended] Use flow-on-PCA synthetic cells as a stand-in for real cells (flow_pca_s0)

- [not_recommended] Share or release the synthetic cells outside the group that holds the real data (vae_nb_s0)

Summary (agent-written)

Generated report 1: pbmc (cont.)

The objective asked for a low-dimensional representation of the PBMC cells and realistic synthetic cells for sharing. The agent proceeded with a negative-binomial VAE on raw counts of the 2,000 most variable genes (the representation model and primary generator) and a RealNVP flow on 20 PCs of log-normalized data (a generation comparator), two seeds each. The VAE latent space generalizes and keeps neighborhood structure about as well as a same-size PCA, but slightly worse (trustworthiness 0.8073 vs PCA baseline 0.8328 for vae_nb_s0). Neither model produced realistic cells in the joint sense: a classifier separates synthetic from held-out real cells easily (C2ST AUC 0.9457 for vae_nb_s0 and 0.8572 for flow_pca_s0, against a real-vs-real reference of 0.5855). The representation is recommended only conditionally, synthetic cells are not recommended as a stand-in for real cells or for sharing, and there is no formal privacy guarantee.

How to read this report

Run names and seeds

Run names are <model id>_s<seed>: vae_nb_s0 is model vae_nb trained with seed 0. Models in this report: vae_nb (seeds 0, 1); flow_pca (seeds 0, 1).

A seed fixes the model's random initialization, mini-batch order, the validation slice used for early stopping, and the random draws when sampling; the held-out split and preprocessing are the same for every run. Models with two seeds are trained twice as a stability check: a conclusion that holds for both seeds does not depend on one lucky or unlucky initialization. Two runs are far too few to estimate run-to-run variability, so the spread between seeds is not a confidence interval and should not be read as one.

Where realism is measured: the evaluation space

Synthetic and held-out real samples are compared in one shared space: the first 14 principal components (PCs) of the transformed data, fit on the training split. These 14 PCs explain 22.6% of the variance of the transformed training data, so the realism metrics (C2ST, MMD, precision/recall, DCR) say nothing about the remaining 77.4%. Per-feature statistics are computed on all features instead.

Why only 14: PCA fitted to the training data overfits it in later components, which would let a classifier tell training data from new real data apart and bias every comparison. The pipeline keeps the components that hold up on unseen data (rule: largest k where an unseen half of the training data keeps >= 0.85 of the fitted half's cumulative PCA variance).

Reference values

Every realism metric is shown with a real-vs-real reference: the same metric computed between a subsample of the training data and the held-out real data. That is the value two genuinely real samples of this size give, so a model is judged by how close it gets to the reference, not to a theoretical ideal.

How memorization is characterized

For each sample, DCR (distance to closest record) is the distance to its nearest training sample in the evaluation space. The held-out real samples give the yardstick: how close a new sample from the same population typically lands to some training sample. dcr_ratio compares the median synthetic DCR to the median held-out DCR (about 1 = as far as new real data; well below 1 = suspiciously close). too_close_fraction counts synthetic samples closer than the 5th percentile of held-out DCRs (about 0.05 is expected without copying).

This detects near-copies of training samples. It is not a privacy guarantee: it does not test membership inference, attribute inference, or any formal property such as differential privacy.

How the learned representation is judged

Trustworthiness (0 to 1) asks whether samples that are neighbors in the latent space are also neighbors in the full transformed data; it is penalized when the latent space puts unrelated samples together. It is compared with a PCA of the same size, fit on the training data, which is the simplest alternative representation.

How to read this report (cont.)

Caution when comparing flow_pca with the other models

flow_pca is trained on the top 20 PCA components of the transformed data, and realism is scored in a 14-component PCA of the same transformed data. The directions it models directly are largely the directions the realism metrics look at, while vae_nb must reproduce them through a model of all 2000 features. The shared realism metrics may therefore favor it. Per-feature and count-level checks, which use all features, are a fairer comparison of what each model reproduces outside those components.

Objective interpretation

Interpretation

Primary task: representation_learning

Secondary tasks: data_generation

The objective explicitly asks for two things joined by 'and': 'learn a low-dimensional representation' maps to representation_learning, and 'generate realistic synthetic cells for sharing' maps to data_generation. Representation is listed first and is the part a single model (a VAE) delivers directly, so it is the primary task; generation is a secondary task evaluated against held-out real cells.

Success criterion

Representation: the held-out real cells' reconstruction loss is close to the training value and the latent space preserves nearest-neighbor structure (trustworthiness) of the log-normalized held-out data; generation: a classifier cannot tell synthetic from held-out real cells much better than its real-vs-real reference AUC, and synthetic cells are no closer to training cells than held-out real cells are.

Whether to model, and how

Decision: proceed (task actually run: representation_learning)

profile.json reports 2,700 cells (well above the 200-sample floor) with no missing values, and neither task needs labels (has_labels is false, which does not matter here). The data are non-negative integer UMI counts (is_nonneg_integer true) with 97.4% zeros, so a count-likelihood generative model (negative-binomial VAE) is a sound way to get both a latent representation and samples. A non-generative alternative (PCA on log-normalized data) would give a low-dimensional representation but cannot generate count-valued cells, so it is not a substitute for the full objective; PCA is still the evaluation space and the flow comparator is built on it. DATA_DESCRIPTION.md warns that with only 2,700 cells large models can memorize training cells, so memorization checks against held-out cells are part of the success criterion.

Other readings considered

- 'For sharing' could be read as a request for privacy-preserving or anonymized data. This pipeline only checks memorization via nearest-neighbor distances (synthetic-to-train vs held-out-to-train), which is evidence, not a formal privacy guarantee such as differential privacy; the output carries no formal privacy guarantee and is not described as private, anonymized, or safe to share.

- data_generation could have been the primary task with representation as a by-product; it was kept secondary because the objective names the representation first and both are evaluated anyway.

- 'Realistic' is not defined; it is interpreted as distributional similarity to held-out real cells in a common PCA space of log-normalized expression (C2ST, MMD), not as preservation of any specific biological signal such as cell-type proportions, which cannot be checked because the dataset has no labels.

Data and preprocessing

Dataset (from DATA_DESCRIPTION.md)

Peripheral blood mononuclear cells (PBMCs) from a healthy donor, profiled with 10x Genomics single-cell RNA sequencing (scRNA-seq). This is the "pbmc3k" dataset used throughout the Scanpy tutorials and widely used as a benchmark for single-cell analysis methods. Stored as the raw h5ad file from the Scanpy tutorials.

Profile

2,700 samples x 32,738 features; modality tabular

Sparsity 0.9741; value range [0, 419]; non-negative integers: True

Labels: none

Held-out split

test_fraction 0.2, stratified: False -> 2160 training / 540 held-out samples

Reason: With 2,700 cells, 0.2 holds out 540 real cells for the two-sample tests (C2ST, MMD) and nearest-neighbor memorization checks while keeping 2,160 cells for training. stratify is false because profile.json reports has_labels: false, so there is no class mix to preserve.

Preprocessing (applied identically for every model; fit on training data only)

- select_top_variance {"n_features": 2000}: profile.json reports 32,738 genes with 97.4% zeros; most genes are almost never detected in 2,700 cells and would add noise and training time. Keeping the 2,000 most variable genes (ranked on library-size-normalized, log-transformed training cells only) keeps the informative genes. 2,000 is a judgment call; 1,000 or 5,000 were alternatives, and 2,000 balances signal against model size for 2,160 training cells.

- normalize_total : DATA_DESCRIPTION.md notes total counts per cell vary about 10x between cells; normalizing library size makes the transformed view and the PCA evaluation space reflect expression composition rather than sequencing depth.

- log1p : Counts are heavy-tailed (profile max 419 against a mean of 0.072); log1p compresses the few highly expressed genes so they do not dominate distances in the PCA evaluation space. The NB-VAE still trains on raw counts, unaffected by this step.

Evaluation space

All models are scored in one shared space: PCA of the transformed data with 14 components (0.2258 of training variance), fit on the training split. The number of components is the largest k where an unseen half of the training data keeps >= 0.85 of the fitted half's cumulative PCA variance: a PCA with more components overfits the training data, and a classifier could then tell training from held-out real samples apart, biasing every comparison.

Models, hyperparameters, and training

vae_nb

Family: vae; input: raw; seeds [0, 1]; serves representation_learning, data_generation; status planned; estimated 2.0 min/seed

Why this model: The only implemented family with a latent space, so it is the primary model for the representation task, and it also samples cells for the generation task. Raw UMI counts (is_nonneg_integer, 97.4% sparse) suit an NB likelihood. Two seeds because the budget allows it and claims about the representation and sample fidelity should not rest on one initialization. Compared against the flow on PCA and against the real-vs-real reference in the shared evaluation space.

- latent_dim = 10: n_train is 2,160 cells; a 10-dimensional latent is the representation the objective asks for and is enough for PBMC-scale structure without fitting noise. Judgment call: 16 or 20 were alternatives, but with ~2k cells extra dimensions mostly fit noise.

- hidden_dims = [256, 128]: The input has 2,000 selected genes (>= 1,000 features), so [256, 128] gives enough capacity to compress 2,000 genes to 10 dimensions while staying small relative to 2,160 training cells, which limits memorization risk flagged in DATA_DESCRIPTION.md.

- epochs = 400: 400 is a maximum for 2,160 training cells; early stopping on the 10% validation split (patience 25) will usually stop sooner, and at ~0.9 min to 400 epochs on ~2,000 x 2,000 it fits the 20-minute budget.

- batch_size = 128: 128 gives ~15 steps per epoch on the ~1,944 cells left after the validation split, enough gradient updates per epoch on this dataset.

- lr = 0.001: 1e-3 with Adam; this is a first run on pbmc with no prior loss curve showing oscillation, so the lower 5e-4 is not yet warranted.

- likelihood = nb: profile.json reports non-negative integer UMI counts with 97.4% zeros and ~10x library-size variation; a negative binomial decoder with library-size scaling models overdispersed sparse counts directly on raw counts and generates count-valued cells.

- kl_warmup_epochs = 40: 40 is ~10% of the 400-epoch maximum; ramping the KL weight avoids posterior collapse early, which matters because the latent space is the primary deliverable here.

Training (vae_nb_s0): 91 of 400 max epochs, kept epoch 66 (validation loss did not improve for 25 epochs); 0.174 min on cpu; 1944 fit / 216 validation samples

Training (vae_nb_s1): 98 of 400 max epochs, kept epoch 73 (validation loss did not improve for 25 epochs); 0.185 min on cpu; 1944 fit / 216 validation samples

flow_pca

Family: flow; input: pca; seeds [0, 1]; serves data_generation; status planned; estimated 0.5 min/seed

Why this model: Comparator for the generation task: an exact-likelihood model with a different inductive bias (density on the top 20 PCs of log-normalized expression rather than an NB decoder on raw counts), scored in the same PCA evaluation space. It has no latent

Models, hyperparameters, and training (cont.)

representation, so it does not serve representation_learning. Two seeds since it is cheap on 2,160 cells.

- pca_components = 20: Density estimation needs many samples per dimension; with 2,160 training cells, 20 components keeps ~100 cells per dimension. The flow models the top 20 PCs of log-normalized expression and treats the remaining variance as independent Gaussian noise when generating; the variance kept is recorded in the run sidecar. Judgment call: 10 would be safer statistically but would drop more PBMC structure.

- n_layers = 8: 8 affine coupling layers on a 20-dimensional input give enough mixing for multi-modal (multiple cell populations) structure while training in ~0.2 min on ~2,000 cells.

- hidden_dim = 128: 128 because there are only 2,160 training cells (well under 20k); a larger coupling network would overfit this small set.

- epochs = 400: 400 is a maximum for 2,160 cells; early stopping on the validation split ends it sooner, and the timing table puts the full 400 epochs at ~0.2 min.

- batch_size = 128: 128 gives ~15 updates per epoch on ~1,944 non-validation training cells, matching the VAE so the comparison isn't confounded by batch size.

- lr = 0.001: 1e-3 with Adam; first run on pbmc, no oscillating loss curve has been observed that would justify lowering it.

Training (flow_pca_s0): 38 of 400 max epochs, kept epoch 13 (validation loss did not improve for 25 epochs); 0.024 min on cpu; 1944 fit / 216 validation samples; input PCA keeps 0.2396 of variance

Training (flow_pca_s1): 39 of 400 max epochs, kept epoch 14 (validation loss did not improve for 25 epochs); 0.024 min on cpu; 1944 fit / 216 validation samples; input PCA keeps 0.2396 of variance

Evaluation focus

Representation: train vs held-out reconstruction loss gap and latent trustworthiness against log-normalized held-out cells. Generation: C2ST AUC and MMD of synthetic vs held-out real cells in the shared PCA space, each judged against its real-vs-real reference, plus nearest-neighbor distance ratios to training cells as memorization evidence (not a privacy guarantee), which matters because the objective mentions sharing.

Evaluation metrics (held-out real data)

Realism metrics use the 14-component evaluation space (22.6% of transformed-data variance); per-feature statistics use all features. Each realism metric has a real-vs-real reference row.

Fidelity

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
n_compared	540	540	540	540
mmd2	0.01961	0.01639	0.00767	0.00959
mmd2_reference	0.00026	-0.00028	0.00026	-0.00028
c2st_auc	0.9457	0.9253	0.8572	0.8352
c2st_auc_reference	0.5855	0.5664	0.5855	0.5664
precision	0.7926	0.8722	0.5611	0.5315
recall	0.7667	0.7667	0.987	0.9833

Memorization

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
dcr_median_synthetic	3.0228	2.8673	3.2329	3.1514
dcr_median_heldout	2.029	2.029	2.029	2.029
dcr_ratio	1.4897	1.4131	1.5933	1.5531
too_close_fraction	0.0056	0.0019	0	0.0074

Evidence about copying of training samples, not a formal privacy guarantee.

Feature stats

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
mean_corr	0.9978	0.9976	0.998	0.9976
var_corr	0.8745	0.8912	0.9694	0.9718
zero_frac_corr	0.9919	0.9916	n/a	n/a
zero_frac_synthetic	0.735	0.7357	n/a	n/a
zero_frac_heldout	0.7232	0.7232	n/a	n/a
library_median_ratio	0.9984	0.9896	n/a	n/a

Likelihood

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
train_mean	-1563.406	-1561.089	-25.368	-25.274
heldout_mean	-1586.174	-1585.719	-24.616	-24.54
per_dimension_heldout	-0.7931	-0.7929	-1.2308	-1.227

Likelihoods are in each model's own input space and units; do not compare them across models with different input spaces.

Representation

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
latent_dim	10	10	n/a	n/a
trustworthiness	0.8073	0.7917	n/a	n/a
trustworthiness_pca_baseline	0.8328	0.8327	n/a	n/a
knn_overlap	0.16	0.1475	n/a	n/a

Evaluation metrics (held-out real data) (cont.)

knn_overlap_pca_baseline	0.1681	0.1677	n/a	n/a

Neighborhoods are Euclidean in the transformed space, which favors PCA (it is the best linear Euclidean summary by construction); a latent close to the baseline is not a failure.

How to read the metrics

- mmd2: Squared maximum mean discrepancy (RBF kernel, bandwidth = median distance among held-out real samples) between synthetic and held-out real samples in the evaluation PCA space. 0 = same distribution; compare with mmd2_reference. The estimator is unbiased, so values near 0 can be slightly negative.

- mmd2_reference: Same MMD, but a training-set subsample vs held-out real: the value two real samples of this size give.

- c2st_auc: Classifier two-sample test: cross-validated ROC AUC of a gradient-boosted classifier separating synthetic from held-out real samples. 0.5 = indistinguishable, 1.0 = perfectly separable.

- c2st_auc_reference: Same test, training subsample vs held-out real (should be near 0.5).

- precision: Fraction of synthetic samples inside the 5-NN radius of some held-out real sample (fidelity: are samples realistic?).

- recall: Fraction of held-out real samples inside the 5-NN radius of some synthetic sample (coverage: is the data's variety reproduced?).

- dcr_median_synthetic: DCR = distance to closest record: for one sample, the Euclidean distance (evaluation space) to its nearest training sample. This is the median DCR over synthetic samples.

- dcr_median_heldout: Median DCR over held-out real samples: how close a genuinely new sample from the same population typically is to some training sample. This is the yardstick for the synthetic DCR.

- dcr_ratio: dcr_median_synthetic / dcr_median_heldout. About 1 means synthetic samples are as far from the training data as new real samples are; well below 1 means they sit unusually close to training samples (possible copying); above 1 means they are farther away than real data (often a sign of off-distribution samples, not of privacy).

- too_close_fraction: Fraction of synthetic samples whose DCR is below the 5th percentile of held-out DCRs. If the model does not copy, about 0.05 is expected (the same share as for new real samples); much higher means a group of synthetic samples are near-copies of training samples. Evidence about copying only, not a privacy guarantee.

- mean_corr: Pearson correlation, across features, between per-feature means of synthetic and held-out real data (transformed space).

- var_corr: Same, for per-feature variances.

- zero_frac_corr: Same, for each feature's fraction of exact zeros (raw counts; only for models that generate counts).

- library_median_ratio: Median total count per synthetic sample divided by the held-out median (counts only).

- train_mean: Mean per-sample ELBO (VAE, a lower bound) or exact log-likelihood (flow) on training data, in the model's own input space. Not comparable across models with different input spaces.

How to read the metrics (cont.)

- heldout_mean: Same quantity on held-out real data; a large drop from train_mean indicates overfitting.

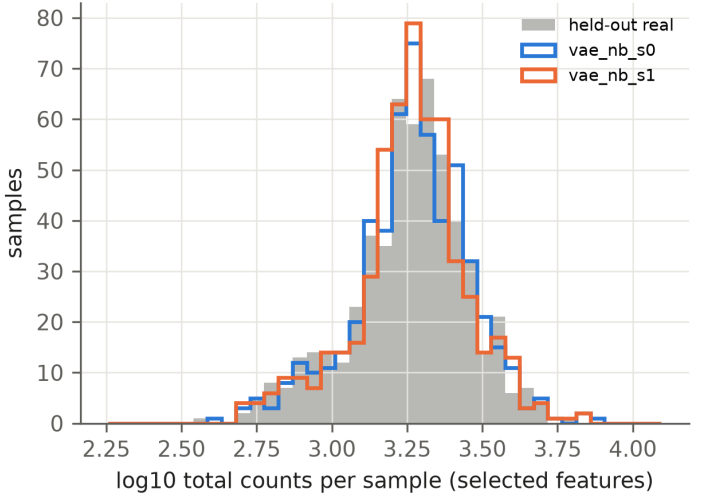
- trustworthiness: Trustworthiness (Venna and Kaski, 2001), between 0 and 1: for each held-out sample, look at its 15 nearest neighbors in the latent space and penalize any that are NOT among its 15 nearest neighbors in the reference space (full transformed data), more heavily the farther away they really are. 1 = every latent-space neighbor is a true neighbor; lower = the latent space puts unrelated samples next to each other.

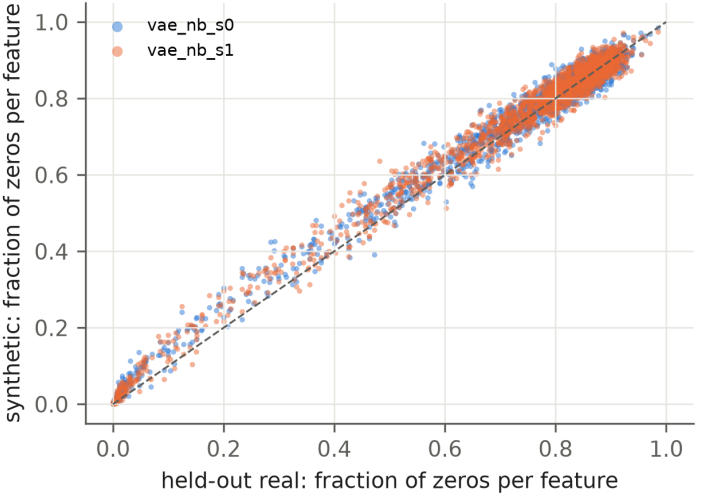
- trustworthiness_pca_baseline: Trustworthiness of a PCA (fit on training data) with as many dimensions as the latent space: the bar a learned representation is compared with.

- knn_overlap: Average fraction of each held-out sample's 15 nearest neighbors in the full transformed space that are also among its 15 nearest neighbors in the latent space.

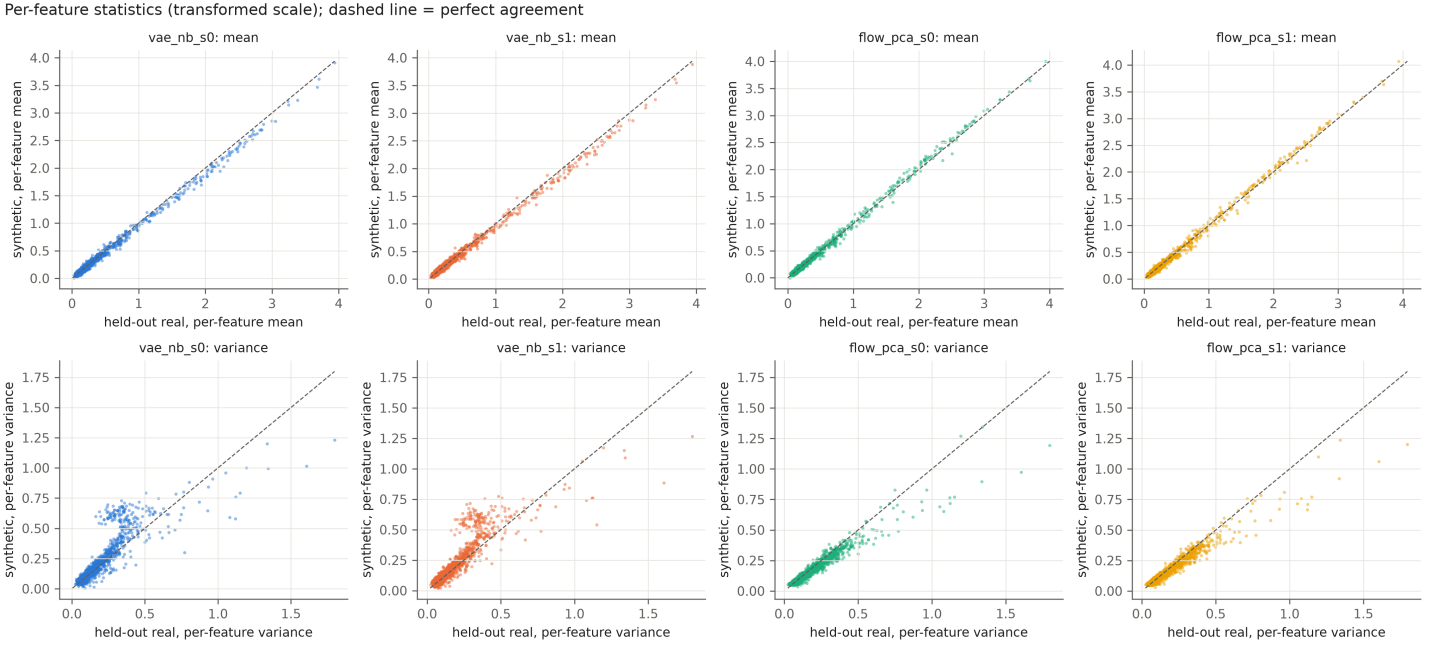
- knn_overlap_pca_baseline: Same, for the same-size PCA baseline.

Count-level properties

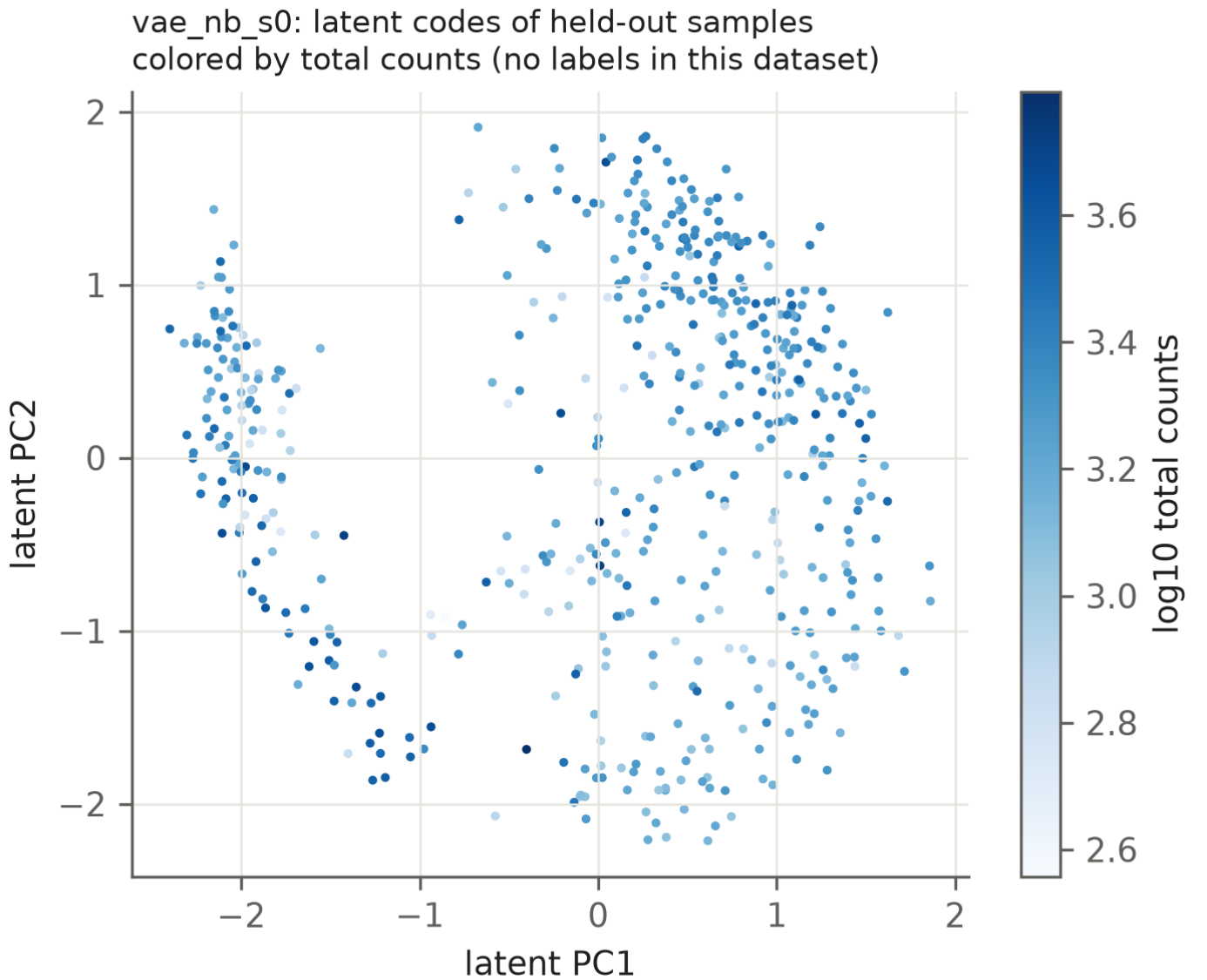




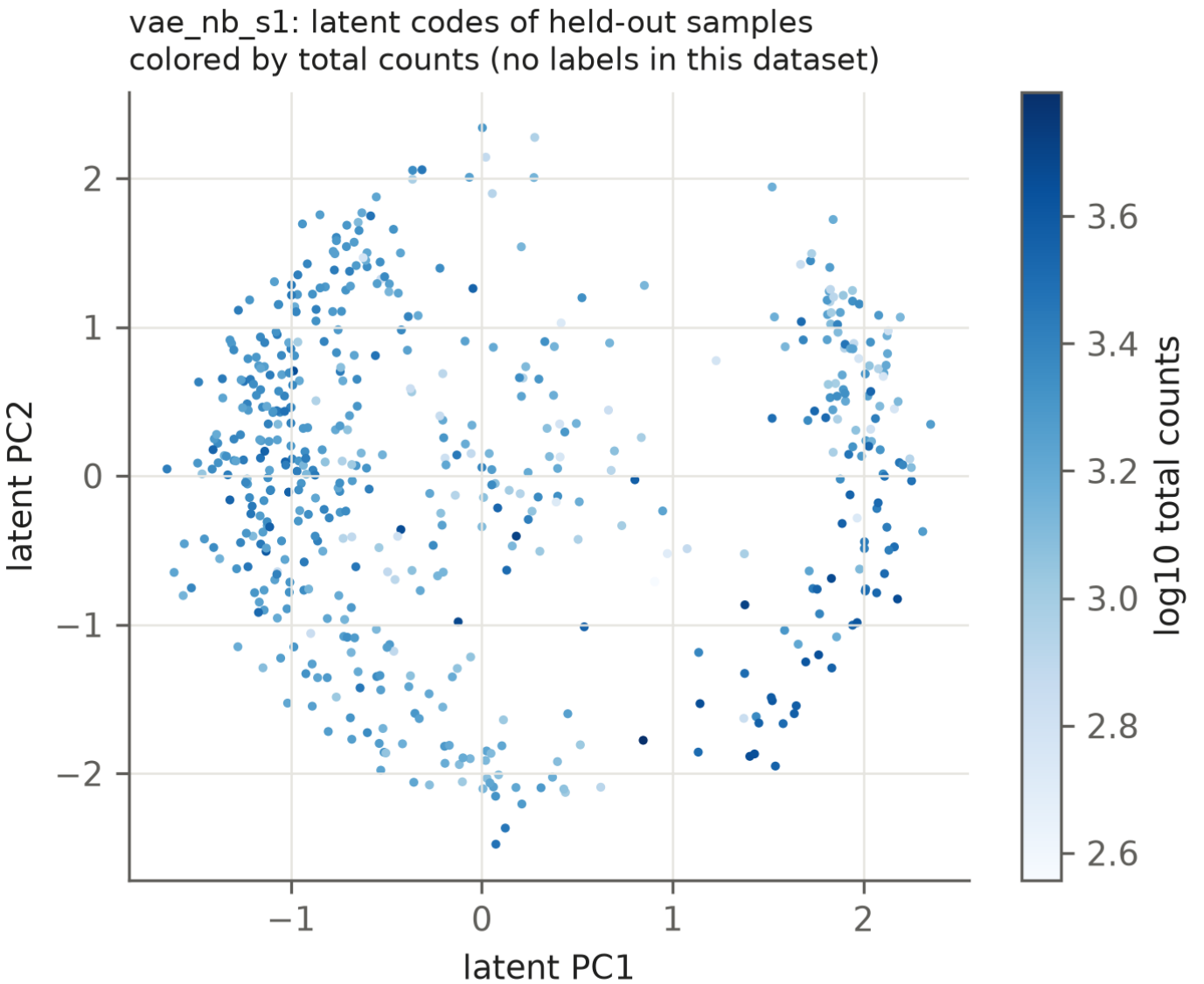
Count-level checks for models that generate counts: total counts per sample, and per-feature fraction of zeros.



Per-feature mean and variance of synthetic vs held-out real data on the transformed scale. Points on the dashed line agree exactly.

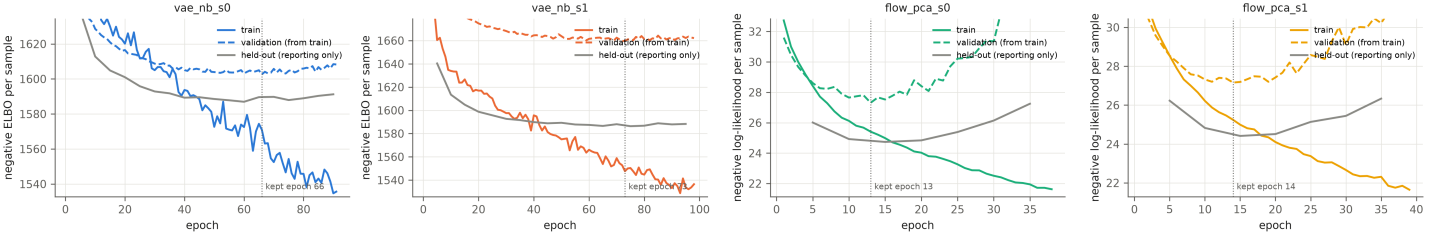


Two-dimensional PCA of a VAE's latent codes for held-out samples. A picture for orientation; neighbor-preservation numbers are on the metrics page.

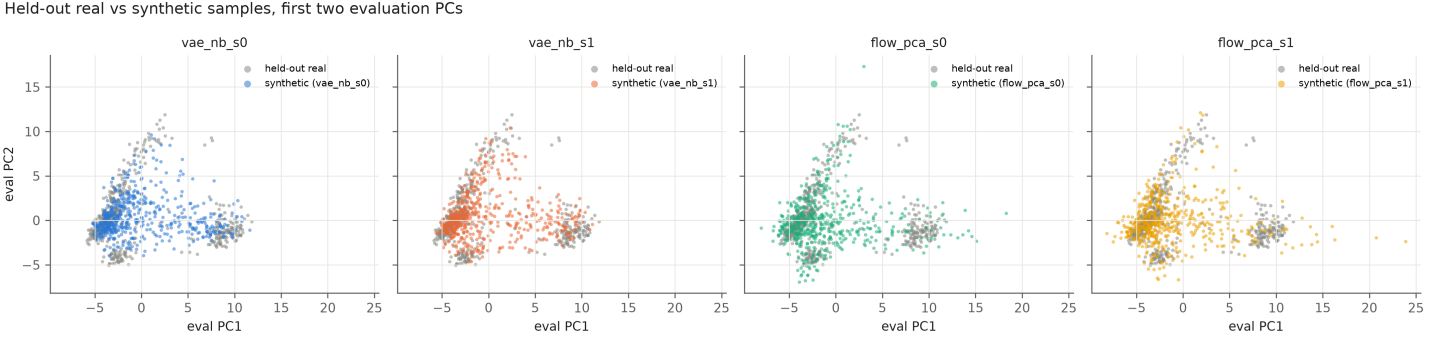


Two-dimensional PCA of a VAE's latent codes for held-out samples. A picture for orientation; neighbor-preservation numbers are on the metrics page.

Training curves



Training curves. The validation set is 10% of the training split and decides early stopping; the held-out curve is shown for reference only and never used to choose anything.



Held-out real samples (gray) and synthetic samples in the first two components of the shared evaluation space. Overlap in two components is necessary but not sufficient; the C2ST and MMD metrics use all components.

Findings

What each model produced

vae_nb_s0 (NB-VAE, seed 0). C2ST AUC 0.9457 (reference 0.5855); MMD² 0.01961 (reference 0.00026); precision 0.7926, recall 0.7667. Per-gene means (correlation 0.9978), zero fractions (0.9919) and median library size (ratio 0.9984) match held-out cells closely; per-gene variances less so (0.8745). Memorization: DCR ratio 1.4897, too-close fraction 0.0056. Held-out ELBO -1586.174 vs training -1563.406. Representation: trustworthiness 0.8073 (PCA baseline 0.8328), kNN overlap 0.16 (PCA baseline 0.1681). Training stopped early at epoch 91 because validation loss had not improved for 25 epochs; epoch 66 was kept.

vae_nb_s1 (seed 1, stability check). C2ST AUC 0.9253 (reference 0.5664); MMD² 0.01639 (reference -0.00028); precision 0.8722, recall 0.7667. DCR ratio 1.4131, too-close fraction 0.0019. Trustworthiness 0.7917 (PCA baseline 0.8327), kNN overlap 0.1475 (PCA baseline 0.1677). Stopped at epoch 98, epoch 73 kept. The conclusions from seed 0 hold for seed 1.

flow_pca_s0 (RealNVP on 20 PCs, which keep 0.2396 of the variance; the rest is added back as Gaussian noise). C2ST AUC 0.8572 (reference 0.5855); MMD² 0.00767 (reference 0.00026); precision 0.5611, recall 0.987. Per-gene mean correlation 0.998, variance correlation 0.9694. DCR ratio 1.5933, too-close fraction 0.0. Stopped at epoch 38, epoch 13 kept, with validation loss rising after that.

flow_pca_s1 (seed 1). C2ST AUC 0.8352 (reference 0.5664); MMD² 0.00959 (reference -0.00028); precision 0.5315, recall 0.9833. DCR ratio 1.5531, too-close fraction 0.0074. Stopped at epoch 39, epoch 14 kept. Same conclusions as seed 0.

All four runs finished; the whole run took 1.19 minutes of the 20-minute budget.

What the figures show

Real vs synthetic (first two evaluation PCs): held-out real cells form a few compact groups (one large group on the left, an arm going up, a separate group on the right). VAE samples cover the same overall range but spread into the empty space between the real groups instead of sitting in them. Flow samples follow the real groups more closely but also scatter between them and include outliers well beyond the real range on PC1.

Count properties: the VAE's per-cell library-size histograms overlap the held-out histogram, and per-gene zero fractions lie close to the diagonal, with synthetic cells having slightly more zeros for genes that are rarely zero in real cells.

Per-feature statistics: per-gene means lie on the diagonal for all four runs. For the VAE, a band of genes with real variance around 0.25 to 0.5 has inflated synthetic variance, and the highest-variance genes are under-dispersed; the flow's variances track the diagonal more closely except for the highest-variance genes.

Latent space (held-out cells, colored by total counts): both VAE seeds place held-out cells into distinct groups (one separate group and a larger connected region). Library size does not visibly drive the layout. Without labels, the groups cannot be named; this is exploratory only.

Training curves: VAE validation and held-out loss flatten after about 40 to 60 epochs while training loss keeps falling, so early stopping chose a reasonable point. Flow validation

Findings (cont.)

loss turns upward after about 13 epochs, a sign of overfitting on the small training set.

How the models compare

On metrics computed in the same 14-component evaluation space, the flow is closer to the real data than the VAE on C2ST and MMD and covers far more of the real variety (recall 0.987 vs 0.7667 for seed 0), but fewer of its samples are realistic (precision 0.5611 vs 0.7926). The flow is trained on PCA scores of the same log-normalized data that defines the evaluation space, which may favor it on these realism metrics. The VAE is the only model that generates UMI counts and matches count-level properties (zero fractions, library size), and the only one with a latent representation. Neither model shows signs of copying training cells. Likelihoods are not comparable across the two models: the VAE's ELBO is on 2,000 raw-count genes and the flow's exact log-likelihood is on 20 PCA dimensions.

For representation: the VAE. For count-shaped synthetic data with realistic per-gene marginals: the VAE. Neither for realistic joint structure.

Answer to the objective

Partly. The representation part is met in a limited sense: the VAE's held-out ELBO (-1586.174) is close to its training value (-1563.406), and the latent space keeps neighborhood structure, but it does not beat a same-size PCA (trustworthiness 0.8073 vs 0.8328). The generation part is not met: C2ST AUC is far above its real-vs-real reference for every run (best 0.8352 vs 0.5664 for flow_pca_s1), so the synthetic cells are not realistic in their joint structure. The memorization criterion is met (synthetic cells are no closer to training cells than held-out real cells are; DCR ratio 1.4897 for vae_nb_s0), but that is evidence about copying, not a privacy guarantee, and it does not make the data suitable for sharing.

Downstream use of the generated data

Use the NB-VAE 10-dimensional latent codes as a low-dimensional representation of these

Verdict: CONDITIONAL | task: representation_learning | model: vae_nb_s0

The latent space generalizes (held-out ELBO is close to training ELBO) and keeps neighborhood structure of held-out cells, and the latent plot shows separated groups of cells, but on both seeds it is slightly below a same-size PCA of log-normalized data on trustworthiness and kNN overlap. For neighbor-based uses a 10-component PCA does about as well; what the VAE adds is a count-aware (negative binomial) model whose latent space is tied to a generator.

- evidence: vae_nb_s0.representation.trustworthiness = 0.8073 (vs vae_nb_s0.representation.trustworthiness_pca_baseline = 0.8328)

- evidence: vae_nb_s0.representation.knn_overlap = 0.16 (vs vae_nb_s0.representation.knn_overlap_pca_baseline = 0.1681)

- evidence: vae_nb_s1.representation.trustworthiness = 0.7917 (vs vae_nb_s1.representation.trustworthiness_pca_baseline = 0.8327)

- evidence: vae_nb_s0.likelihood.heldout_mean = -1586.174 (vs vae_nb_s0.likelihood.train_mean = -1563.406)

- caveat: Use it if a count-likelihood representation or a generator sharing the same latent space is wanted; otherwise PCA on log-normalized data is an equally good and simpler representation by these metrics.

- caveat: The neighborhood metrics are Euclidean in the log-normalized space, which favors PCA by construction.

- caveat: No cell-type labels exist, so whether latent groups correspond to cell types is not tested.

- caveat: Raw data were not QC-filtered; low-quality cells or doublets are embedded along with everything else.

Use NB-VAE synthetic cells as a stand-in for real PBMC cells in analyses that depend on th

Verdict: NOT_RECOMMENDED | task: data_generation | model: vae_nb_s0

A classifier separates VAE synthetic cells from held-out real cells easily on both seeds (AUC far above the real-vs-real reference), and MMD is far above the real-vs-real value; the PCA scatter shows synthetic cells filling the gaps between real cell populations instead of matching their compact clusters.

- evidence: vae_nb_s0.fidelity.c2st_auc = 0.9457 (vs vae_nb_s0.fidelity.c2st_auc_reference = 0.5855)

- evidence: vae_nb_s0.fidelity.mmd2 = 0.01961 (vs vae_nb_s0.fidelity.mmd2_reference = 0.00026)

- evidence: vae_nb_s1.fidelity.c2st_auc = 0.9253 (vs vae_nb_s1.fidelity.c2st_auc_reference = 0.5664)

Downstream use of the generated data (cont.)

- caveat: Results could change with a different architecture or longer training; only this configuration was tested.

Use NB-VAE synthetic count matrices where only per-gene and per-cell marginal propertie

Verdict: CONDITIONAL | task: data_generation | model: vae_nb_s0

Per-gene means, per-gene zero fractions, and library sizes of the synthetic counts closely track held-out real cells, so the output looks like a PBMC count matrix gene by gene; per-gene variances match less well (a band of genes has inflated variance in the feature-stats figure).

- evidence: vae_nb_s0.feature_stats.mean_corr = 0.9978 (vs perfect agreement = 1.0)

- evidence: vae_nb_s0.feature_stats.zero_frac_corr = 0.9919 (vs perfect agreement = 1.0)

- evidence: vae_nb_s0.feature_stats.library_median_ratio = 0.9984 (vs held-out real = 1.0)

- evidence: vae_nb_s0.feature_stats.var_corr = 0.8745 (vs flow_pca_s0.feature_stats.var_corr = 0.9694)

- caveat: Only marginal properties are reproduced; the joint structure is not (see the C2ST result above).

- caveat: Any use outside the original data holders is subject to the privacy statement: no formal privacy guarantee.

Use flow-on-PCA synthetic cells as a stand-in for real cells

Verdict: NOT_RECOMMENDED | task: data_generation | model: flow_pca_s0

The flow is closer to real data than the VAE on C2ST and MMD and covers the real variety (high recall), but it is still clearly separable from real cells, and fewer than 6 in 10 of its samples lie near a real cell (low precision), with outliers far outside the real range in the PCA scatter. It models only 20 PCs of log-normalized data and produces no counts.

- evidence: flow_pca_s0.fidelity.c2st_auc = 0.8572 (vs flow_pca_s0.fidelity.c2st_auc_reference = 0.5855)

- evidence: flow_pca_s0.fidelity.precision = 0.5611 (vs vae_nb_s0.fidelity.precision = 0.7926)

- evidence: flow_pca_s0.fidelity.recall = 0.987 (vs vae_nb_s0.fidelity.recall = 0.7667)

- evidence: flow_pca_s1.fidelity.c2st_auc = 0.8352 (vs flow_pca_s1.fidelity.c2st_auc_reference = 0.5664)

- caveat: The flow's samples are log-normalized expression, not UMI counts, so they cannot feed count-based tools.

- caveat: The flow early-stopped within about 15 epochs with a rising validation loss, so its fit is limited by the small training set.

Share or release the synthetic cells outside the group that holds the real data

Downstream use of the generated data (cont.)

Verdict: NOT_RECOMMENDED | task: data_generation | model: vae_nb_s0

The memorization check found no sign of copying (synthetic cells are farther from training cells than new real cells are, and few are unusually close), but that is not a privacy guarantee, and the synthetic cells are not realistic enough in their joint structure to serve the 'realistic cells for sharing' goal anyway.

- evidence: vae_nb_s0.memorization.dcr_ratio = 1.4897 (vs about 1 for new real cells (dcr_median_heldout = 2.029))
- evidence: vae_nb_s0.memorization.too_close_fraction = 0.0056 (vs about 0.05 expected for new real cells)
- evidence: vae_nb_s0.fidelity.c2st_auc = 0.9457 (vs vae_nb_s0.fidelity.c2st_auc_reference = 0.5855)
- caveat: Any future sharing would need a proper privacy review or a formally private training method (e.g. differential privacy); this pipeline offers neither.
- caveat: The DCR ratio above 1 mainly reflects off-distribution samples, not protection.

Not tested (no recommendation either way)

- Whether latent clusters or synthetic cells reproduce PBMC cell-type identities and proportions: the dataset has no labels; testing it would need annotated cells (e.g. marker-based annotation) and a per-type comparison.
- Whether a downstream model (e.g. a classifier or clustering) trained on synthetic cells performs as well on real held-out cells as one trained on real cells (train-on-synthetic, test-on-real).
- Resistance to membership-inference or other privacy attacks: only nearest-neighbor distances were measured.
- Gene-gene correlation structure beyond what the 14-component evaluation PCA (22.6% of variance) captures.

Privacy

There is no formal privacy guarantee (such as differential privacy) for any output of this run. The only check was memorization: the median distance from each synthetic cell to its nearest training cell, compared with the same distance for held-out real cells, in a 14-component PCA space. For the NB-VAE (seed 0) the ratio was 1.4897 and 0.0056 of synthetic cells fell below the held-out 5th-percentile distance (about 0.05 expected); for the flow (seed 0) the ratio was 1.5933 with 0.0 too close. This is evidence that the models did not copy individual training cells in that space; it does not show that information about individual donors or cells cannot be inferred, and it does not cover the remaining variance outside the evaluation space.

Limitations

- Models trained on PCA scores are evaluated in a PCA space of the same data, which may favor them on the shared realism metrics (see 'How to read this report').
- Fidelity is measured in a 14-component PCA space (0.2258 of training variance); differences outside it are not measured.
- No independent critique was run for this report.
- The data were not QC-filtered (the description warns of possible low-quality cells and doublets); no row filtering step was available, so these cells were modeled and evaluated with the rest.
- The dataset has no labels, so neither cell-type structure in the latent space nor cell-type proportions in synthetic data could be checked.
- The evaluation space is a 14-component PCA of log-normalized data; fidelity and memorization are judged only in that space.
- The flow overfit quickly (best epoch 13 of 38) on 1,944 fitting cells; a smaller flow or fewer PCs might help, but was not tried in this run.
- No critic agent was available in this pipeline, so no independent critique was applied.
- There is no formal privacy guarantee. The nearest-neighbor memorization check is evidence about copying of training cells only.

Reproducibility

Command: ./run.sh pbmc "Learn a low-dimensional representation and generate realistic synthetic cells for sharing" --report-number 1

Plan seed 0; plan revision 1; budget 20 min; pipeline time 1.19 min

Preprocessing hash 93b023f05bc08e80

python 3.13.5, torch 2.14.0, numpy 2.5.3, scikit-learn 1.9.1

- vae_nb_s0: seed 0, config hash 1e3e55c97da71f7c, 0.174 min on cpu

- vae_nb_s1: seed 1, config hash 4184940d0ff3beed, 0.185 min on cpu

- flow_pca_s0: seed 0, config hash 95a7adb96933bcfd, 0.024 min on cpu

- flow_pca_s1: seed 1, config hash cee5bfb809e3621c, 0.024 min on cpu

Generated 2026-10-23 16:25. All intermediate files: outputs/pbmc/.